

GEOMETRIC HÖLDER THEORY FOR KÄHLER METRIC SPACES

BIN GUO, JIAN SONG, JACOB STURM

ABSTRACT. In this paper, we initiate the study of geometric Hölder regularity for singular Kähler spaces. We prove that Hölder continuity of the Kähler potential implies Hölder continuity of the distance function induced by the corresponding Kähler current on a normal Kähler space. As a consequence, we obtain geometric Hölder equivalence among intrinsic metric structures from canonical metrics and extrinsic metric structures from algebraic embeddings.

1. INTRODUCTION

Theorem 1.1. *Let (X, θ) be a compact normal Kähler variety of complex dimension n equipped with a smooth Kähler metric. Suppose $\varphi \in \text{PSH}(X, \omega_0) \cap C^\alpha(X) \cap C^2(X_{\text{reg}})$ for some $\alpha \in (0, 1)$ and a smooth non-negative closed $(1, 1)$ -form ω_0 such*

$$\omega = \omega_0 + \partial\bar{\partial}\varphi$$

is a Kähler metric on X_{reg} . Then there exists $\delta = \delta(\alpha) > 0$ and $C = C(\alpha, \theta, \|\omega_0\|_{C^0(X)}, \|\varphi\|_{C^\alpha(X)}, \delta) > 0$ such that for any $p, q \in X_{\text{reg}}$, we have

$$d_\omega(p, q) \leq C (d_\theta(p, q))^\delta.$$

Although theorem 1.1 is stated globally for a compact normal Kähler variety, it can be localized and hold for any bounded open subset of an analytic variety with normal singularities (after shrinking the domain suitably). This is because the proof of Theorem 1.1 is essentially local.

Obviously, (X_{reg}, ω) is a smooth Kähler manifold. We let $(\mathbf{X}, \mathbf{d})$ be the metric completion of (X_{reg}, ω) . Naturally, one would like to understand the topological, holomorphic and metric structures of $(\mathbf{X}, \mathbf{d})$.

Corollary 1.1. *Under the same assumption of Theorem 1.1, $(\mathbf{X}, \mathbf{d})$ is a compact metric space. Furthermore, if ω is a Kähler current, i.e., there exists $c > 0$ such that*

$$\omega \geq c\omega_0,$$

then $(\mathbf{X}, \mathbf{d})$ is a compact metric space homeomorphic to X itself.

Theorem 1.2. *Let (X, θ) be a compact normal Kähler variety with klt singularities equipped with a smooth Kähler metric θ . Suppose ω is an admissible Kähler current such that the metric space (X, d_ω) is a non-collapsed $\text{RCD}(-\lambda, 2n)$ -space. Then there*

exist $\delta = \delta(\theta, [\omega], p, \|\frac{\omega^n}{\theta^n}\|_{L^p(X, \theta^n)}, \lambda) > 0$, $c = c(\theta, [\omega], p, \|\frac{\omega^n}{\theta^n}\|_{L^p(X, \theta^n)}, \lambda) > 0$ and $C = C(\theta, [\omega], p, \|\frac{\omega^n}{\theta^n}\|_{L^p(X, \theta^n)}, \lambda, \delta) > 0$ such that

$$cd_\theta(p, q) \leq d_\omega(p, q) \leq C (d_\theta(p, q))^\delta$$

for all $p, q \in X$.

Corollary 1.2. *Let (X, θ) be a compact normal Kahler variety with klt singularities equipped with a smooth Kahler metric θ . Suppose ω_1 and ω_2 are two admissible Kahler current such that both metric spaces (X, d_{ω_1}) and (X, d_{ω_2}) are non-collapsed $RCD(-\lambda, 2n)$ -space. Then there exist $\delta' = \delta'(\theta, [\omega], p, \|\frac{\omega^n}{\theta^n}\|_{L^p(X, \theta^n)}, \lambda) > 0$ and $C' = C'(\theta, [\omega], p, \|\frac{\omega^n}{\theta^n}\|_{L^p(X, \theta^n)}, \lambda, \delta') > 0$ such that*

$$(C')^{-1} (d_{\omega_1}(p, q))^{1/\delta'} \leq d_{\omega_2}(p, q) \leq C (d_{\omega_1}(p, q))^{\delta'}$$

for all $p, q \in X$. Similar Holder equivalence in Corollary ?? holds.

Suppose (X, θ) is a compact normal Kahler variety with klt singularities that satisfies one of the following

- (1) $\dim X = 3$;
- (2) X is locally smoothable;
- (3) X admits a crepant resolution.

For any admissible singular Kahler metric ω whose Ricci current is bounded below by $-\lambda$ for some $\lambda \geq 0$, (X, ω) can be identified as a non-collapsed $RCD(-\lambda, 2n)$ -space. Therefore, local Kahler potentials of ω are indeed Holder continuous and the conclusions in Theorem 1.2 and Corollary 1.2 automatically hold.

We now consider the case where the underlying Kahler spaces are no longer fixed. Let

$$\mathcal{K}(n, D)$$

be the set of n -dimensional compact Kahler manifold (X, ω) equipped with a smooth Kahler metric ω satisfying the following condition

$$(1.1) \quad \text{Ric}(\omega) \geq -\omega,$$

$$(1.2) \quad \text{Diam}(X, \omega) \leq D,$$

$$(1.3) \quad [\omega] \in H^{1,1}(X, \mathbb{R}) \cap H^2(X, \mathbb{Z}).$$

There exists $N = N(n, D)$ such that every (X, ω) can be embedded in $\mathbb{CP}^N$ by the orthonormal basis with respect to the L^2 -inner product induced by a sufficiently large power of $[\omega]$. Let ω_{FS} be the Fubini-Study metric of $\mathbb{CP}^N$. Then the restriction $\omega_{FS}|_X$ to X induces a singular Kahler metric as well as an intrinsic distance function $d_{\omega_{FS}|_X}$ on X .

Theorem 1.3. *There exist $\delta = \delta(n, D) > 0$ and $C = C(n, D) > 0$ such that for any $(X, \omega) \in \mathcal{K}(n, D)$ and any $p, q \in X$,*

$$C^{-1} d_{\omega_{FS}|_X}(p, q) \leq d_\omega(p, q) \leq C (d_{\omega_{FS}|_X}(p, q))^\delta.$$

2. HOLDER ESTIMATES IN THE SMOOTH CASE

A uniform Hölder estimate for distance functions is established by Li [Li] for smooth Kähler metrics with uniform L^p ($p > 1$) volume measure on a fixed Kähler manifold. Li's theorem can be reduced to the following form due to Kolodziej's Hölder estimate for complex Monge-Ampère equations.

Proposition 2.1. *Let (X, ω_0) be a compact Kähler manifold (or a bounded domain in $\mathbb{C}^n$) equipped with a smooth Kähler metric ω_0 . If $\varphi \in \text{PSH}(X, \omega_0) \cap C^2(X)$, then for any $\alpha \in (0, 1)$, there exists $C = C(X, \omega_0, \alpha, \|\varphi\|_{C^\alpha(X, \omega_0)}) > 0$ such that for all $p, q \in X$,*

$$(2.4) \quad d_{\omega_\varphi}(p, q) \leq C d_{\omega_0}(p, q)^{\alpha/2},$$

where $\omega_\varphi = \omega_0 + \partial\bar{\partial}\varphi$.

The key ingredients are Kolodziej's Hölder estimate for complex Monge-Ampère equations and the scale invariant Poincaré inequality for smooth Riemannian metrics. Kolodziej's Hölder estimate can be generalized to the singular case when X is a compact Kähler space with normal singularities without much modification. However, the scale invariant Poincaré inequality does not appear to be available in the singular case, particularly for normal compact Kähler varieties equipped with a smooth Kähler metric. Naturally one can propose the following question.

Problem 2.1. *Let X be a compact normal variety holomorphically embedded in a smooth Kähler manifold $(\mathcal{M}, g_{\mathcal{M}})$. Is $(X, g_{\mathcal{M}}|_X)$ an PI space, i.e., a metric measure space that satisfies*

- (1) *the volume doubling condition,*
- (2) *a scale invariant Poincaré inequality?*

Here $g_{\mathcal{M}}|_X$ is the restriction of $g_{\mathcal{M}}$ to X , which induces a distance function on X by taking the metric completion of $g_{\mathcal{M}}|_X$ on the smooth part of X . It is well-known that X satisfies the volume doubling condition with respect to the extrinsic distance function. However, the intrinsic distance poses more challenges, and there are very few tools to establish a local scale-invariant Poincaré inequality.

In this section, we aim

- (1) to provide a new proof for Theorem 2.1 without the use of the local scale-invariant Poincaré inequality,
- (2) and to generalize Theorem 2.1 by assuming $\varphi \in C^2(X \setminus S) \cap C^\alpha(X)$ for some analytic subvariety S of X , where X is still assumed to be smooth.

More precisely, we will prove the following theorem.

In Theorem ?? below, we introduce a different approach which allows us to treat singular metrics and also provides refined geometric information, even in the smooth setting.

Theorem 2.1 says that $d_\varphi(p, q) \leq Ad_0(p, q)^{\alpha/2}$. In other words, for every $p, q \in X$, if γ is the ω_0 geodesic joining p to q , there is another (non-unique) curve η joining p to q such that

$$(2.5) \quad \ell_\varphi(\eta) \leq C\ell_0(\gamma)^{\alpha/2}$$

One may naturally ask for the relationship between the curves η and γ . For example, can we choose $\eta = \gamma$? In the following, we show that this can be done, up to an ε error.

We first give a slight generalization of Theorem ???. Let (X, ω_0) be a smooth Kähler manifold and let $\mathcal{S}$ be an analytic subvariety of X .

Proposition 2.2. *Let (X, ω_0) be a compact Kähler manifold. Suppose $\varphi \in \text{PSH}(X, \omega_0) \cap C^2(X \setminus \mathcal{S})$ satisfies the uniform Hölder continuity*

$$|\varphi(p) - \varphi(q)| \leq Kd_0(p, q)^\alpha$$

for all $p, q \in X$. Then there exists $C > 0$, depending on (X, ω_0) and K, α , such that for any $p, q \in X \setminus \mathcal{S}$, there exists a piecewise smooth curve $\gamma \subset X \setminus \mathcal{S}$ joining p and q with

$$d_\varphi(p, q) \leq C(d_0(p, q))^{\alpha/2}.$$

Proof. As in the statement of the theorem, we shall use $C > 0$ to denote a constant depending on (X, ω_0) and K, α (but independent of ε). We allow the constant C to change value from line to line to avoid introducing new constants $C_1, C_2, \dots$ etc.

Once we fix (X, ω_0) , we can work in local charts with holomorphic coordinates so that

$$C^{-1}\omega_{\text{euc}} \leq \omega_0 \leq C\omega_{\text{euc}},$$

where ω_{euc} is the euclidean metric on $\mathbb{C}^n$. Note that

$$(2.6) \quad \omega_\varphi = \omega_0 + \sqrt{-1}\partial\bar{\partial}\varphi = \omega_{\text{euc}} + \sqrt{-1}\partial\bar{\partial}(\varphi - \psi)$$

where ψ is a local potential for ω_0 . Since $\|\psi\|_{C^\alpha} \leq C$, we will write φ instead of $\varphi - \psi$ so that in local coordinates, $\omega_\varphi = \omega_{\text{euc}} + \sqrt{-1}\partial\bar{\partial}\varphi$.

We may assume $p = (0, 0, \dots, 0) \in \mathbb{C}$. Let $\tilde{\mathcal{S}}$ be the set of point $q \in B(0, 1) \subset \mathbb{C}$ such that the straight line segment in $\mathbb{C}$ joining p and q does not intersect $\mathcal{S}$. Obviously, $\tilde{\mathcal{S}}$ contains $\mathcal{S}$ and $\tilde{\mathcal{S}}$ is a closed subset of $\mathcal{S}$ and has dimension no greater than $2n - 1$.

For simplicity, we assume $q = (d, 0, \dots, 0) \in \tilde{\mathcal{S}}$ with $d > 0$. We let γ be the straight line segment joining p to q , which is a ω_{euc} geodesic satisfying

$$C^{-1}d_0(p, q) \leq |q| = \ell_{\text{euc}}(\gamma) \leq Cd_0(p, q).$$

and let $z = (d/2, 0, \dots, 0)$ be the midpoint of γ .

For $x \in B_r(z)$, the euclidean ball of radius r centered at z in $\mathbb{C}^n$, we define $\eta_{1,x} : [0, 1] \rightarrow \mathbb{C}^n$ to be the straightline line segment joining p and x , i.e., $\eta_{1,x}(t) = tx$. This

is the euclidean segment joining p to x . Similarly, we let $\eta_{2,x}$ to be the euclidean segment joining x to q . We then define

$$\eta_x = \eta_{1,x} \cup \eta_{2,x}$$

be the piece-wise straight line joining p and q through x , which avoids $\mathcal{S}$ by the choice of $q \in \tilde{\mathcal{S}}$. We wish to show that for an appropriately chosen value of x that η and γ satisfy (??).

Obviously, we have

$$\begin{aligned} \ell_\varphi(\eta_{1,x}) &= \int_0^1 (g_{i\bar{j}} x^i x^{\bar{j}})^{1/2} dt \\ &\leq \int_0^1 ((g_{\text{euc}})_{i\bar{j}} x^i x^{\bar{j}} \cdot \text{tr}_{g_{\text{euc}}}(g))^{1/2} dt \\ &= |x| \int_0^1 \left[n + (\Delta_{g_{\text{euc}}} \varphi)(tx) \right]^{1/2} dt. \end{aligned}$$

Let $\rho(x) \in C_c^\infty(B_{2r}(z))$ be the usual cutoff function which equals one on $B_r(z)$ and let dx denote Euclidean measure, and let $V_r = \text{vol}_{\text{euc}}(B_r(z))$. Then

$$\begin{aligned} \int_{B_r(z) \setminus \tilde{\mathcal{S}}} \ell_\varphi(\eta_{1,x}) dx &\leq \int_{B_{2r}(z) \setminus \tilde{\mathcal{S}}} \ell_\varphi(\eta_{1,x}) \rho(x)^2 dx \\ &\leq d \int_{B_{2r}(m)} \int_0^1 \left(\left(n + (\Delta_{\text{euc}} \varphi)(tx) \right) \rho(x) \right)^{1/2} dt dx \\ &\leq d \cdot V_{2r}^{1/2} \int_0^1 \left(\int_{B_{2r}} \left(n + (\Delta_{\text{euc}} \varphi)(tx) \right) \rho(x) dx \right)^{1/2} dt \\ &= d \cdot V_{2r}^{1/2} \int_0^1 \left(\int_{B_{2r}} \left(n + \frac{1}{t^2} \Delta_{\text{euc}} \left(\varphi(tx) - \varphi(0) \right) \right) \rho(x) dx \right)^{1/2} dt \\ &= d \cdot V_{2r}^{1/2} \int_0^1 \left(\int_{B_{2r}} \left(n \rho(x) + \frac{1}{t^2} \cdot (\varphi(tx) - \varphi(0)) \Delta \rho(x) \right) dx \right)^{1/2} dt. \end{aligned}$$

By the estimates of $\Delta \rho$, we have

$$\begin{aligned}
\int_{B_r(z)} \ell_\varphi(\eta_{1,x}) dx &\leq Cd \cdot V_{2r}^{1/2} \int_0^1 \left(nV_{2r} + \int_{B_{2r}} \frac{1}{t^2} |\varphi(tx) - \varphi(0)| \frac{1}{r^2} dx \right)^{1/2} dt \\
&\leq Cd \cdot V_{2r}^{1/2} \int_0^1 \left(V_{2r}^{1/2} + \left(\int_{B_{2r}} \frac{1}{t^2} |\varphi(tx) - \varphi(0)| \frac{1}{r^2} dx \right)^{1/2} \right) dt \\
&\leq Cd \cdot V_{2r} + Cd \cdot V_{2r}^{1/2} \cdot K^{1/2} \cdot \frac{1}{r} \int_0^1 \frac{1}{t} \left(\int_{B_{2r}} |tx|^\alpha dx \right)^{1/2} dt \\
&\leq Cd \cdot V_{2r} + Cd \cdot V_{2r}^{1/2} \cdot \frac{1}{r} \int_0^1 t^{\alpha/2-1} \left(\int_{B_{2r}} |x|^\alpha dx \right)^{1/2} dt \\
&\leq CV_r d^{\alpha/2}.
\end{aligned}$$

The same inequality holds if we replace $\eta_{1,x}$ by $\eta_{2,x}$. Adding the two, we obtain

$$\frac{1}{V_r} \int_{B_r(z)} \ell_\varphi(\eta_x) dx \leq Cd^{\alpha/2} = C(d_0(p, q))^{\alpha/2}.$$

Therefore there exists a piecewise smooth curve γ' joining p and q satisfying

$$\ell_\varphi(\gamma') \leq C(d_0(p, q))^{\alpha/2}.$$

□

We now give an alternative proof of Proposition 2.2 using the segment inequality. It is potentially useful for a quantitative approach. In particular, the dependence of constant $C > 0$ in Proposition 2.2 on (X, ω_0) can be made more precise, which depends on the diameter bound and Ricci lower bound of (X, ω_0) .

Alternative proof of Proposition 2.2.

□

In this note we apply our previous work [2] on Hölder continuity of singular Kähler metrics to obtain geometric estimates for singular Kähler metrics on smoothable varieties. We show that Hölder continuity of the potential on a singular variety W implies Hölder continuity of the distance function. As a corollary, the metric completion of the smooth locus $W^{\text{reg}} \subseteq W$ is homeomorphic to W . Here is a typical application.

Theorem 2.1. *Let (X, ω_{KE}) be a klt Kähler-Einstein variety, and let ω_0 be a smooth background metric, and let d_{KE} and d_0 be the corresponding distance functions. Assume X is smoothable. Then there exists $C > 0$ and $\alpha > 0$ such that*

$$\frac{1}{C} \cdot d_0 \leq d_{\text{KE}} \leq C \cdot d_0^\alpha$$

We next show how the technique use in the proof can be used to prove the following.

Theorem 2.2. *Let $W \subseteq \mathbb{P}^N$ be normal variety $\varphi \in \text{PSH}(W, \omega_0) \cap C^\alpha(W) \cap C^\infty(W^{\text{reg}})$. Here $\omega_0 := \omega_{\text{FS}}|_W$ and we set $\omega_\varphi = \omega_0 + \sqrt{-1} \partial \bar{\partial} \varphi > 0$. Assume $\text{diam}_{\omega_\varphi}(W^{\text{reg}}) < \infty$. Let d_0, d_{ω_φ} be the distance functions on W^{reg} associated to ω_0 and ω_φ respectively. Assume $\omega_\varphi \geq \frac{1}{C} \omega_0$ for some $C > 0$. Then there exists $\delta > 0$ and $C > 0$ such that for all $x, x' \in W^{\text{reg}}$,*

$$\frac{1}{C} d_0(x, x') \leq d_{\omega_\varphi}(x, x') \leq C d_0^\delta(x, x')$$

In particular, the metric completion of $(W^{\text{reg}}, d_{\omega_\varphi})$ is homeomorphic to W .

Here $d_{\omega_\varphi}(x', x) = \inf \{ \int_\gamma |\gamma'(t)|_{\omega_\varphi} dt : \gamma : [0, 1] \rightarrow W^{\text{reg}}, \gamma(0) = x, \gamma(1) = x', \}$.

Remark: We state the theorem here for normal varieties, but the proof work as well in the case of normal Kähler spaces.

3. GROWTH OF ALGEBRAIC FUNCTIONS

In order to prove Theorem 2.2 we shall need some preliminary results on growth of algebraic functions.

Let $\pi : X \rightarrow Z$ be a covering space and $\gamma : [0, 1] \rightarrow Z$ a continuous curve. Let $z = \gamma(0)$, $z' = \gamma(1)$, and $x \in X_z := \pi^{-1}(z)$. The the “lift of γ to x ” is the unique curve $\lambda : [0, 1] \rightarrow X$ such that $\lambda(0) = x$ and $\pi \circ \lambda = \gamma$. In particular, $\lambda(1) = x'$ for some $x' \in \pi^{-1}(z')$.

Let $B_r(p) = \{z \in \mathbb{C} : |z - p| < r\}$, let $d \geq 1$ and let $a_1, \dots, a_d$ be holomorphic functions on $B_R(0)^n \subseteq \mathbb{C}^n$, where $R = 4d(4d + 4)^{d-1}$. Assume $\sup_{B_R^n(0)} |a_j| < 1$ for $1 \leq j \leq d$ and let

$$f(z, x) = x^d + \sum_{j=1}^d a_j(z) x^{d-j}$$

Note that by Fujiwara's bound, for a fixed $z \in B_R^n(0)$, if $f(z, x) = 0$ then $|x| < 2$. Let $B \subseteq B_R(0)^n$ be the subvariety consisting of those points $z \in B_R(0)^n$ defined by the property $|\{x \in \mathbb{C} : f(z, x) = 0\}| < d$. Here the roots of the polynomial $f(z, x) = 0$ are counted without multiplicity. Let $X' = \{(z, x) \in B_1(0)^n \times B_2(0) : f(z, x) = 0\}$ and let $\pi : X' \rightarrow B_1(0)^n$ be the map $(z, x) \mapsto z$. Let $Z = B_1(0)^n \setminus B$ and $X = \pi^{-1}(Z)$. Then $\pi : X \rightarrow Z$ is a covering map of degree d .

Lemma 3.1. *Let $a_1, \dots, a_d$ be holomorphic functions as above, and define $f(z, x)$ and B as above. Let $z', z \in Z$ and assume the curve $\gamma : [0, 1] \rightarrow Z$ has the properties*

- (1) $\gamma(0) = z, \gamma(1) = z'$
- (2) $\gamma(t) \notin B$ for all $t \in [0, 1]$.
- (3) $|\gamma(t) - z| \leq |z' - z|$ for all $t \in [0, 1]$.

Let $x \in \pi^{-1}(z)$ and let λ be the lift of L to z and let $x' = \lambda(1)$. Then

$$(3.1) \quad |\lambda(1) - \lambda(0)| = |x' - x| \leq (2d + 1)|z' - z|^{1/d}$$

Proof. Fix z, z' and x as above and write

$$f(z, x) = \prod_{j=1}^d (x - x_j)$$

with $x = x_1$. We order the x_j such that

$$0 \leq |x_2 - x_1| \leq |x_3 - x_1| \leq \dots \leq |x_d - x_1| < \infty$$

Let $r = |z' - z|$ and $\alpha = r^{1/d}$. Then, by the pigeonhole principle, there exists $1 \leq p \leq 2d + 1$ and $1 \leq j \leq d$ such that

$$|x_j - x_1| \leq (p - 1)\alpha < p\alpha < (p + 1)\alpha \leq |x_{j+1} - x_1|$$

where we define $|x_{d+1} - x_1| := \infty$.

We claim that for each $t \in [0, 1]$, the polynomials $f(z, x), f(\gamma(t), x) \in \mathbb{C}[x]$ have the same number of roots in the ball $B_{p\alpha}(x_1)$. To see this, we use Rouché's theorem: If $|x - x_1| = p\alpha$ then

$$|f(z, x)| = \prod_{j=1}^d |x - x_j| \geq \alpha^d = r$$

On the other hand, if $|x - x_1| = p\alpha$ then $|x| < p\alpha + |x_1| \leq (2d + 1)2 + 2 = 4d + 4$ so

$$|f(\gamma(t), x) - f(z, x)| = \left| \sum_{j=1}^d (a_j(\gamma(t)) - a_j(z)) y^{d-j} \right| < d(4d+4)^{d-1} \max_{1 \leq j \leq d} |a_j(\gamma(t)) - a_j(z)|$$

Fix $t \in (0, 1]$ with $\gamma(t) \neq \gamma(0) = z$ and $\mathbb{C}^1 \subseteq \mathbb{C}^n$ be the complex line containing $\gamma(0)$ and $\gamma(t)$ and $B_R(0) \subseteq \mathbb{C}^1$ the ball of radius R . Then

$$|a_j(\gamma(t)) - a_j(z)| = \left| \frac{1}{2\pi i} \int_{\partial B_R(0)} \frac{a_j(\zeta)(\gamma(t) - z)}{(\zeta - z)(\zeta - \gamma(t))} d\zeta \right| \leq R|z' - z| \frac{1}{(R-1)^2} < \frac{4r}{R}$$

Thus $|f(\gamma(t), y) - f(z, x)| < d(4d+4)^{d-1} \frac{4r}{R} = r$.

Rouché's theorem now implies that $\lambda(t) \in B_{pr^{1/d}}(x)$ for all t . In particular, setting $t = 1$, we obtain

$$|\lambda(1) - \lambda(0)| = |x' - x| \leq pr^{1/d} \leq (2d+1)r^{1/d} = (2d+1)|z' - z|^{1/d} \quad \square$$

Remark: If we choose R to be any number $R > 1$ and assume $|a_j(z)| \leq A$ on $B_R(0)$, then a scaling argument shows that (3.1) still holds with the constant $(2d+1)$ replaced by a constant depending on d, A and R .

4. BI-HÖLDER CONTINUITY OF THE DISTANCE FUNCTION

In this section we prove Theorem 2.2. First we recall the definition of a pluri-subharmonic function on a singular variety. We say $\varphi \in \text{PSH}(X, \omega_0) \iff \omega_0 + \sqrt{-1}\partial\bar{\partial}\varphi = \Phi|_X$ where Φ is a PSH function on an open set in $\mathbb{C}^M$ and $\Phi|_X$ is the restriction of Φ to a local embedding of X in $\mathbb{C}^M$.

Proof of Theorem 2.2. The problem is local so if we let $\mathbb{C}^N \subseteq \mathbb{P}^N$ be an affine open set, we can, for the purpose of proving the theorem, replace W by $W \cap B_R(0)^N$ for some $R > 0$. By the Noether normalization theorem, if $\mathbb{C}^n \subseteq \mathbb{C}^N$ is a generic vector subspace then $\pi : X \rightarrow \mathbb{C}^n$ is a finite map in the sense of algebraic geometry. In particular, π is proper and if d is the degree of π , then the fibers of π contain exactly d points counted with multiplicity. For example, if $X = \{x_1, x_2 \in \mathbb{C} : x_1 x_2 = 1\}$ then we can take $\mathbb{C}^1$ to be any line other than $x_1 = 0$ or $x_2 = 0$. A point in $\mathbb{C}^N$ will be labelled $(z_1, \dots, z_n, w_1, \dots, w_{N-n})$ so π is the projection onto the first n coordinates.

Let $Y = B_R(0)^n$ and let $B \subseteq Y$ be a subvariety containing the image of the singular set as well as the branch locus. Let $Z = Y \setminus B$ and $X = \pi^{-1}(Z) \subseteq \mathbb{C}^N$. Then $\pi|_X : X \rightarrow Z$ is a covering space whose degree we denote by d .

Recall that $\omega_0 = \omega_{\text{FS}}|_W$ and $\omega_\varphi = \omega_0 + \sqrt{-1}\partial\bar{\partial}\varphi$. Let

$$\eta_0 = \pi_*\omega_0 \quad \text{and} \quad \eta_\psi = \pi_*\omega_\varphi = \eta_0 + \sqrt{-1}\partial\bar{\partial}\psi \quad \text{and} \quad \omega_\psi = \pi^*\eta_\psi = \omega_0 + \sqrt{-1}\partial\bar{\partial}(\psi \circ \pi)$$

Then η_0, η_ψ are positive $(1, 1)$ currents on Y which are smooth Kähler metrics on Z , and ω_ψ is a positive $(1, 1)$ current which is smooth on X . Moreover, for $z \in Z$ we have

$$(4.1) \quad \psi(z) = \sum_{j=1}^d \varphi(x_j) \quad \text{where} \quad \pi^{-1}(z) = \{x_1, \dots, x_d\}$$

If $z \in Y$, then (4.1) still holds if we count the points in $\pi^{-1}(z)$ with multiplicity. In particular, η_0 and η_ψ are positive $(1, 1)$ currents on $Y = B_R(0)^n$ with continuous local potentials whose restrictions to $Z \subseteq Y = B_R(0)^n$ are smooth Kähler metrics. In the next lemma we show that the continuous function $\psi : Y \rightarrow \mathbb{R}$ is in fact Hölder continuous.

Lemma 4.1. *There exist constants $C > 0$ and $\tau > 0$ such that for every $z, z' \in Y$ we have*

$$(4.2) \quad |\psi(z') - \psi(z)| \leq C|z' - z|^\tau$$

Proof. We use the notation $\ell_{\eta_\psi}(\gamma)$ to denote the length of a smooth curve γ on X with respect to the η_ψ metric. If $\gamma : [0, 1] \rightarrow Z$ is smooth then for every $1 \leq p \leq n$ we have

$$(4.3) \quad \max_{1 \leq j \leq d} \ell_{\omega_\varphi}(\lambda_j) \leq \ell_{\eta_\psi}(\gamma) \leq \sum_{j=1}^d \ell_{\omega_\varphi}(\lambda_j) \leq \sqrt{d} \ell_{\eta_\psi}(\gamma).$$

Here the $\{\lambda_1, \dots, \lambda_d\}$ are the lifts of γ to the points $x_1, \dots, x_d$. In particular, $\ell_{\omega_\psi}(\lambda_p)$ is independent of p and $\omega_\psi > \omega_\varphi$. The identities (4.3) still hold if z, z' lie in Y by taking limits. For example, if $z \in Y$ then the sequence $x_1, \dots, x_d$ may not consist of distinct elements. For example, if $x_1 = x_2 = x_3$ is a point of multiplicity three, then $\lambda_1, \lambda_2, \lambda_3$ will all emanate from x_1 .

Now let $\gamma(t)$ be a curve satisfying the hypotheses of Lemma 3.1. Let $z = \gamma(0)$ and $z' = \gamma(1)$. Order the set $\pi^{-1}(z') = \{x'_1, \dots, x'_d\}$ so that x_j, x'_j are the endpoints of λ_j . Since φ is Hölder continuous, there exist $C > 0$ and $\alpha \in (0, 1)$ so that $|\varphi(x') - \varphi(x)| \leq Cd_0(x', x)^\alpha$ for all $x, x' \in X$. Then Theorem 4.1 (see below) and Lemma 3.1 (applied to each coordinate w_j for $1 \leq j \leq N - n$) imply

$$(4.4) \quad |\psi(z') - \psi(z)| = \left| \sum_{j=1}^d \varphi(x'_j) - \sum_{j=1}^d \varphi(x_j) \right| \leq \sum_{j=1}^d |\varphi(x'_j) - \varphi(x_j)| \leq \sum_{j=1}^d Cd_0(x'_j, x_j)^\alpha$$

$$(4.5) \quad \leq \sum_{j=1}^d C |x'_j - x_j|^{\beta\alpha} \leq C(2d+1) |z' - z|^{\beta\alpha/d} \quad \square$$

Here we have made use of the following corollary of a theorem of Łojasiewicz [5], which compares the intrinsic euclidean metric with the extrinsic euclidean metric.

Theorem 4.1. *Let $Y \subseteq \mathbb{C}^N$ be an algebraic variety and assume $Y \cap B_1(0)^N$ is connected. Let ω the euclidean metric on $\mathbb{C}^N$ and ω_0 its restriction to Y . Then there exists $C > 0$ and $\beta \in (0, 1)$ such that for all $x, x' \in Y \cap B_1(0)^N$ we have*

$$d_0(x, x') := d_{\omega_0}(x, x') \leq C |x' - x|^\beta$$

Since ψ is continuous we see that

$$|\psi(z') - \psi(z)| \leq C |z' - z|^{\beta\alpha/d}$$

for all $z, z' \in Y$.

Lemma 4.2. *There exists $C > 0$ and $\tau > 0$ such that*

$$(4.6) \quad d_{\eta_\psi}(z', z) \leq C |z' - z|^\tau$$

for all $z, z' \in Z' = Y \setminus S$ where $S = \pi(W^{\text{sing}})$.

Proof. The proof is the same as that of Theorem 2, but η_ψ is not smooth so we must be a little careful. There are two steps that need to be justified, namely (2.6) and (??).

In Theorem 2 the background metric ω_0 is smooth and hence has finite Hölder norm in any C^α . In our setting, the background metric is $\eta_0 = \pi_*\omega_0$.

Now $\omega_0 = \sqrt{-1}\partial\bar{\partial}|x|^2$ so according to (4.1), $\omega_0 = \sqrt{-1}\partial\bar{\partial}\psi_0$ where

$$\psi_0(z) = \sum_{j=1}^d |x_j|^2$$

so (4.5) implies ψ_0 has finite C^α norm for some α , which is all we need to justify (2.6).

To justify the integration by parts step (??), let $\Delta = \{t \in \mathbb{C} : |t| < 1\}$ be the unit disk. Then the function $\Psi(t, x) : \Delta \times Y \rightarrow \mathbb{R}$ given by $\Psi(t, x) := \psi(tx)$ is quasi-psh, since it is the pullback of a quasi-psh function $\psi(z)$ by a holomorphic map. This implies that for fixed $t \in [0, 1] \subseteq \Delta$, that the function $\varphi(tx) : Y \rightarrow \mathbb{R}$ is quasi-psh and bounded. Hence we can integrate by parts.

The inequality (4.6) is nearly what we want; we just want to lift this estimate back up to an estimate on X , i.e. we want to use (4.6) to show that

$$(4.7) \quad d_{\omega_\varphi} \leq C d_{\text{Euc}}^\delta$$

for some $\delta > 0$. It suffices to show

$$(4.8) \quad d_{\omega_\psi}(x_0, x_1) \leq C d_{\text{Euc}}(x_0, x_1)^\delta$$

where $\omega_\psi = \pi^* \pi_* \omega_\varphi$ (since $\omega_\psi > \omega_\varphi$). Here d_{Euc} is the euclidean metric on $X \cap B_1(0)$ and by abuse of notation, we use the notation ψ to denote $\psi \circ \pi$.

Proof of Theorem 2.2. Let $x_0, x_1 \in X$ and $x_t = \lambda(t)$ a curve joining x_0 to x_1 such that $\ell_{\omega_0}(\lambda) \leq d_{\omega_0}(x_0, x_1) + \varepsilon$. We call λ an (ω_0, ε) -geodesic.

Let $z_j = \pi(x_j)$, and $z(t) = \pi \circ \lambda(t)$. Then $z(t)$ joins z_0 to z_1 and clearly $z(t)$ lifts to $\lambda(t)$. For $s < t$, let $\lambda[s, t]$ (*resp.* $z[s, t]$) be λ (*resp.* z) restricted to the interval $[s, t] \cap [0, 1]$

For $1 \leq k \leq d$ we define $S_k \in [0, 1]$ as follows: $t \in S_k$ if there exist a sequence of $(\eta_\psi, 1/j)$ geodesics $\xi_j[0, t] : [0, t] \rightarrow Z$ joining z_0 to $z_t = \pi(x_t)$ with the following property: the closed curve $\gamma[0, t]^{-1} \circ \xi_j[0, t]$ lifts to a curve in X joining x_0^1 to x_0^k .

Observe that $\sup S_x \in S_x$. To see this, choose $\varepsilon_m \geq 0$ with $\varepsilon_p \rightarrow 0$ such that $t_k - \varepsilon_p \in S_k$. Then $\gamma[t_k - \varepsilon_p, t_k] \circ \xi_j[0, t_k - \varepsilon_p]$ joins z_0 to z_k and has length approaching $d_{\eta_\psi}(z_0, z_k)$.

Let m be a positive integer with $m \leq d$. We define real numbers $0 = t_0 < t_1 < \dots < t_m = 1$ as follows.

Let $t_0 = 0$ and $t_1 = \sup S_1$. If $t_1 = 1$ then we set $m = 1$. If $t_1 < 1$ then, after possibly reordering $x_1, \dots, x_d$, we may assume that $S_2 \cap [t_1, 1]$ has t_1 as an accumulation point. Let $t_2 = \sup S_2$. If $t_2 = 1$ then let $m = 2$. Otherwise, we continue this process until $t_m = 1$.

Fix $1 \leq k \leq m - 1$. Choose $\varepsilon_{k,j} \rightarrow 0$ such that $t_k + \varepsilon_{k,j} \in S_{k+1}$ and choose $\xi_{k,j}[0, t_k]$ and $\xi_{k,j}[0, t_k + \varepsilon_{k,j}]$ to be $(\eta_\psi, 1/j)$ approximate geodesics satisfying:

- (1) They join z_0 to $z(t_k)$ and z_0 to $z(t_k + \varepsilon_{k,j})$ respectively.
- (2) $\gamma[0, t_k]^{-1} \circ \xi_{k,j}[0, t_k]$ lifts to a curve in X joining x_0^1 to x_0^k .
- (3) $\gamma[0, t_k + \varepsilon_{k,j}]^{-1} \circ \xi_{k,j}[0, t_k + \varepsilon_{k,j}]$ lifts to a curve in X joining x_0^1 to x_0^{k+1} (possible since $t_k + \varepsilon_{k,j} \in S_{k+1}$).

Let $\varepsilon_{m,j} = \varepsilon$ be a fixed positive number (independent of m, j). Here j is a large integer to be chosen later. In the following we drop the subscript j if there is no fear of confusion.

In Figure 1, the red dots represent $z(t_0), z(t_1), z(t_2), z(t_3), \dots, z(t_m)$, the blue dots represent $z(t_1 + \varepsilon), z(t_2 + \varepsilon), z(t_3 + \varepsilon) \dots$ and the black dots are the elements of B (and thus don't intersect the dotted curves).

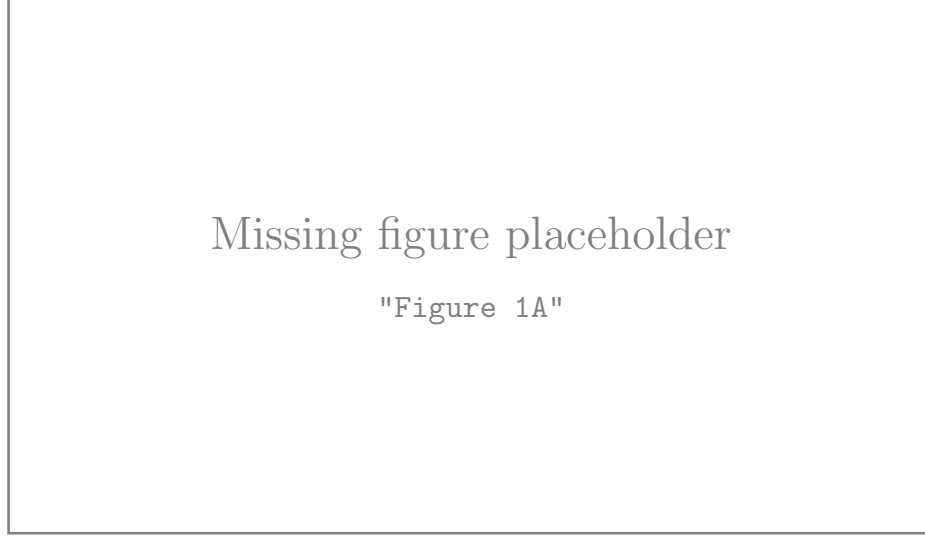


FIGURE 1.

The choice of $\varepsilon = \varepsilon_{k,j}$ implies that the boundaries of the yellow regions in Figure 2 lift to closed curves on X .

Extend $\gamma : [0, 1] \rightarrow Z$ to $\gamma : [0, 1 + \varepsilon_{m,j}] \rightarrow Z$ by setting $\gamma(t) = \gamma(1)$ if $t \geq 1$. Now consider the curve

$$\theta_\varepsilon := \left(\prod_{k=1}^{m-1} \gamma[t_{k+1}, t_{k+1} + \varepsilon_{k,j}] \circ \xi_{k,j}[0, t_{k+1}] \circ \xi_{k,j}[0, t_k + \varepsilon_{1,j}]^{-1} \right) \circ \gamma[t_1, t_1 + \varepsilon_{1,j}] \circ \xi_{1,j}[0, t_1]$$

where $\varepsilon_{k+1} = \varepsilon$ if $k < m - 1$ and 0 if $k = m - 1$. and let Θ_ε be the lift of θ_ε to x_0 . Then

$$\ell_{\omega_\varphi}(\Theta_\varepsilon) \leq \sum_{k=1}^m \left(\ell_{\eta_\psi}(\xi_{k,j}[0, t_k + \varepsilon_{k,j}]) + \ell_{\eta_\psi}(\xi_{k,j}[0, t_{k-1}]) \right) + O(\varepsilon)$$

Now (4.6) implies

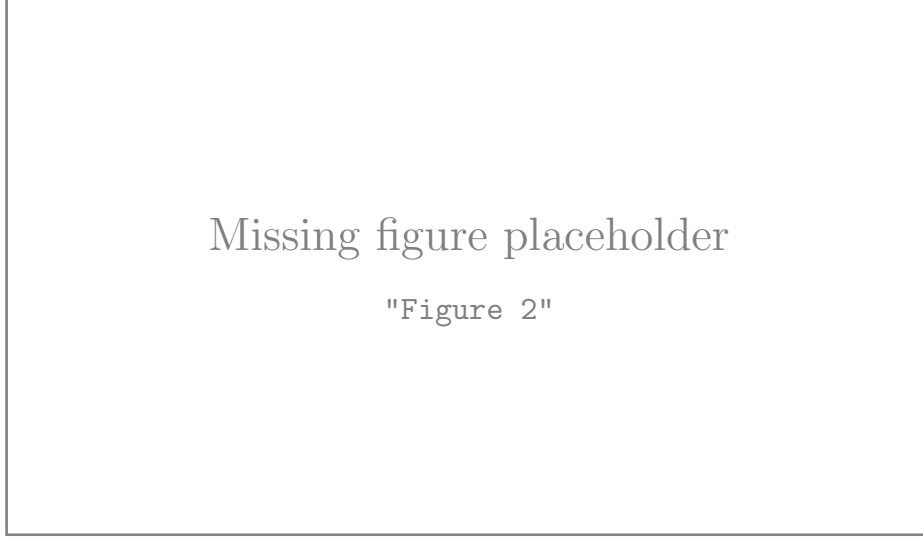


FIGURE 2.

$$\begin{aligned} \ell_{\eta_\psi}(\xi_j[0, t]) - \frac{1}{j} &\leq d_{\eta_\psi}(0, z_t) \leq d(z, z_t)^\alpha \leq d(x, x_t)^\alpha \leq C d_{\omega_0}(x_0, x_t)^\alpha \\ &\leq C \ell_{\omega_0}(\lambda[0, t])^\alpha \leq C d_{\omega_0}(x_0, x_1)^\alpha + \varepsilon \end{aligned}$$

Thus

$$\ell_{\omega_\varphi}(\Theta_\varepsilon) \leq C \cdot 2m \cdot d_{\omega_0}(x_0, x_1)^\alpha + O(\varepsilon)$$

On the other hand, Θ_ε is, by construction, a curve that connects x_0 to x_1 .

$$d_{\omega_\varphi}(x_0, x_1) \leq \ell_{\omega_\varphi}(\Theta) \leq 2mC \cdot d_{\omega_0}(x_0, x_1)^\alpha + O(\varepsilon)$$

Taking the limit as $\varepsilon \rightarrow 0$ we obtain (4.7). □

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